

RAFHA, Saudi Arabia

WMO: 403620

Lat: 29.626N

Lon: 43.491E

Elev: 1474

StdP: 13.93

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.9	37.8	10.6	10.0	52.2	15.9	13.0	55.7	26.6	56.0	23.5	56.2	3.6	300	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	28.1	113.4	69.3	111.5	69.1	109.6	68.7	75.1	105.6	72.9	105.0	71.4	103.3	9.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	96.0	89.6	62.3	88.6	86.7	59.4	79.8	83.9	39.7	105.8	37.6	104.7	36.1	102.7	91.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.5	21.5	18.9	DB	29.3	116.4	4.9	1.9	25.8	117.8	23.0	118.9	20.2	119.9	16.7	121.3	(4)
(5)			WB	25.6	77.0	4.2	6.0	22.6	81.4	20.1	84.9	17.8	88.2	14.8	92.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.5	52.5	56.4	64.9	74.7	85.3	91.5	94.5	95.5	89.8	80.2	64.0	55.0		
	DBStd	16.59	6.01	6.59	7.11	6.78	5.41	3.90	3.78	3.90	4.60	6.10	6.99	6.52		
	HDD50	83	39	17	1	0	0	0	0	0	0	0	2	24		
	HDD65	1152	390	249	90	6	0	0	0	0	0	1	100	316		
	CDD50	9379	116	196	463	742	1094	1246	1381	1411	1193	936	422	179		
	CDD65	4972	1	8	87	298	629	796	916	946	743	472	70	6		
	CDH74	78801	26	113	1029	3615	9412	13491	16037	16685	11862	5831	633	67		
	CDH80	53055	1	24	395	1756	5926	9453	11669	12312	8041	3266	204	8		
(14) Wind	WSAvg	8.5	8.2	8.7	9.5	9.8	9.0	8.7	9.0	7.6	7.4	8.2	7.7	7.6		
(15) Precipitation	PrecAvg	3.0	0.6	0.3	0.5	0.4	0.2	0.0	0.0	0.0	0.0	0.2	0.6	0.3		
	PrecMax	5.2	2.0	1.3	3.8	2.5	1.1	0.1	0.0	0.0	0.0	0.8	2.8	1.8		
	PrecMin	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.5	0.6	0.4	0.7	0.6	0.2	0.0	0.0	0.0	0.0	0.2	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.4	83.8	95.9	102.1	109.1	113.3	116.2	116.2	112.6	105.4	90.7	80.7		
		MCWB	57.0	59.0	63.3	66.2	67.3	67.5	71.2	69.6	69.1	65.8	62.2	57.4		
	2%	DB	71.7	77.1	89.3	97.0	105.6	109.9	113.3	114.0	109.5	100.6	85.9	74.9		
		MCWB	54.8	56.3	60.4	64.3	67.5	68.3	69.3	69.5	68.6	65.6	60.8	55.6		
	5%	DB	67.6	73.0	84.1	93.4	102.5	107.9	111.3	112.1	107.1	98.2	81.0	70.1		
		MCWB	52.8	53.8	58.6	63.7	66.5	67.8	68.7	69.6	67.6	65.3	59.1	54.4		
	10%	DB	63.5	69.2	79.2	89.7	100.2	105.9	109.1	109.7	104.1	94.8	76.9	66.4		
		MCWB	50.8	52.9	57.0	62.1	66.0	67.6	68.1	69.4	66.9	64.0	57.8	52.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.2	63.7	66.2	73.2	74.8	76.5	76.2	77.8	76.1	71.4	67.1	61.7		
		MCDB	71.1	76.6	89.1	89.6	100.0	104.0	108.6	109.2	105.7	95.8	78.7	72.7		
	2%	WB	57.2	59.3	63.0	68.4	71.6	72.7	73.6	74.7	72.6	68.8	64.1	59.0		
		MCDB	67.1	70.5	82.0	88.4	98.0	104.5	108.0	107.2	102.7	93.6	75.5	68.6		
	5%	WB	54.8	56.7	60.4	66.0	69.1	70.6	71.8	72.6	70.0	66.9	62.2	56.7		
		MCDB	63.8	67.7	79.5	87.3	96.2	103.3	106.1	106.9	101.3	91.9	73.7	66.8		
	10%	WB	52.6	54.2	58.2	63.8	67.0	68.7	69.7	70.8	67.9	65.2	60.2	54.4		
		MCDB	61.8	66.2	76.6	84.6	95.5	102.1	104.4	105.9	100.2	89.5	72.8	64.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.7	21.6	23.7	24.4	25.5	27.5	27.8	28.1	27.4	25.0	20.9	19.8		
		MCDBR	24.9	26.8	29.0	28.4	28.6	29.4	29.1	28.8	29.0	28.7	26.6	24.5		
		MCWBR	13.6	13.9	13.9	12.3	12.1	12.3	11.7	11.5	12.2	12.8	12.1	12.6		
		MCDBR	21.6	22.5	26.2	25.6	26.3	28.1	28.4	28.7	28.6	26.2	20.7	20.2		
	5% WB	MCWBR	13.0	13.6	14.0	13.0	13.1	14.0	13.7	14.0	14.3	12.9	10.8	11.7		
(40) Clear-Sky Solar Irradiance	taub		0.411	0.451	0.515	0.610	0.632	0.560	0.531	0.543	0.514	0.540	0.477	0.430		
	taud		1.956	1.852	1.708	1.555	1.531	1.654	1.735	1.715	1.759	1.709	1.824	1.920		
	Ebn at Noon		251	252	245	228	224	239	246	241	242	222	226	236		
	Edh at Noon		49	59	73	89	91	80	74	74	69	68	56	49		
(44) All-Sky Solar Radiation	RadAvg		1194	1494	1833	2081	2294	2551	2460	2294	1999	1587	1225	1091		
	RadStd		69	71	70	73	90	60	75	62	66	62	102	64		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+1.31	N/A	-4.09	+1.91	-1.67	N/A	N/A	-130	+605	+461
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page